

Student Number: _____ Teacher Name: _____

PENRITH SELECTIVE HIGH SCHOOL
MATHEMATICS DEPARTMENT



2024 TERM 3 TRIAL HSC EXAMINATION

YEAR 12 Mathematics Extension 1

General Instructions:

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided with this examination paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- No correction tape or white out allowed.

Total marks: 70

Section I – 10 marks (pages 1–3)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 4–7)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Question Topic	Multiple Choice	Q.11	Q.12	Q.13	Q.14	Total
Combinatorics		b / 2	a / 2			/ 4
Functions	3 / 1	a / 3			a / 2	/ 6
Proof	2 / 1			b / 4	b / 4	/ 9
Trigonometric Functions		e / 5	e / 5			/ 10
Calculus	4, 6, 9, 10 / 4		b, c / 3+2	a, c, d / 4+3+4	c / 4	/ 24
Vectors	1, 7, 8 / 3	c / 2	d / 3		d / 5	/ 13
Binomial Distribution	5 / 1	d / 3				/ 4
Total	/ 10	/ 15	/ 15	/ 15	/ 15	/ 70

Section I

10 marks

Attempt Questions 1–10

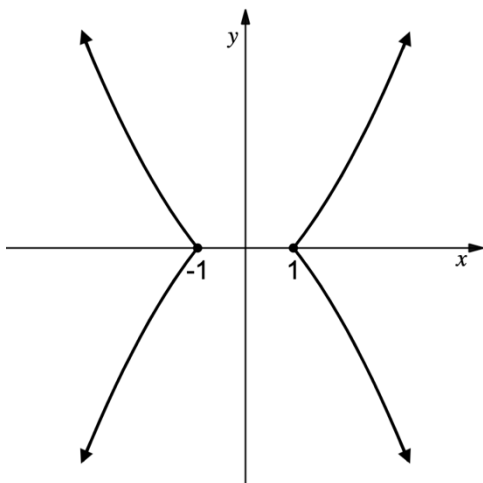
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Given that $\overrightarrow{OA} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$ and $\overrightarrow{OB} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$, What is the direction of $\overrightarrow{AB}$ to the nearest degree?
- A. 37°
 - B. 53°
 - C. -53°
 - D. 143°

- 2 Mathematical induction is a method of proof that can be used to prove a statement:
- A. only for all positive integers n .
 - B. for all integers $n \geq$ any fixed integer.
 - C. for all real numbers n .
 - D. for all integers n .

- 3 For $f(x) = (x + 3)(x - 1)$, which one of the following represents the graph below?



- A. $|y| = f(x)$
- B. $y = |f(|x|)|$
- C. $|y| = |f(x)|$
- D. $|y| = f(|x|)$

4 Which of the following integrals produces the volume of a cone?

A. $\pi \int_0^2 x \, dx$

B. $\pi \int_{-1}^2 (y^2 - 1)^2 \, dy$

C. $\pi \int_{-1}^2 (2 - x)^2 \, dx$

D. $\pi \int_{-2}^0 (2y - 2)^2 \, dy$

5 Find $P(X = 3)$ if $X \sim B\left(4, \frac{5}{8}\right)$.

A. $\binom{4}{3} \left(\frac{5}{8}\right)^3 \left(\frac{3}{8}\right)$

B. $1 - \binom{4}{3} \left(\frac{5}{8}\right) \left(\frac{3}{8}\right)^3$

C. $\left(\frac{5}{8}\right)^3$

D. $\binom{4}{3} \left(\frac{5}{8}\right) \left(\frac{3}{8}\right)^3$

6 Which of the following is **not** a first-order linear differential equation?

A. $xy' + x^2y = 1$

B. $y \sin x - y' \cos x = 0$

C. $y'y + 4x = 2$

D. $y' = 3x^2$

7 A particle is projected with initial speed $V \, \text{ms}^{-1}$ at an angle of α to the horizontal.

If its position at t seconds is given by $\mathbf{r} = 40t\mathbf{i} + \left(9t - \frac{1}{2}gt^2\right)\mathbf{j}$, which of the following statements is correct?

A. $V = 7 \, \text{ms}^{-1}$

B. $V = \frac{\sqrt{6481}}{2} \, \text{ms}^{-1}$

C. $V = \sqrt{1681 - g^2} \, \text{ms}^{-1}$

D. $V = 41 \, \text{ms}^{-1}$

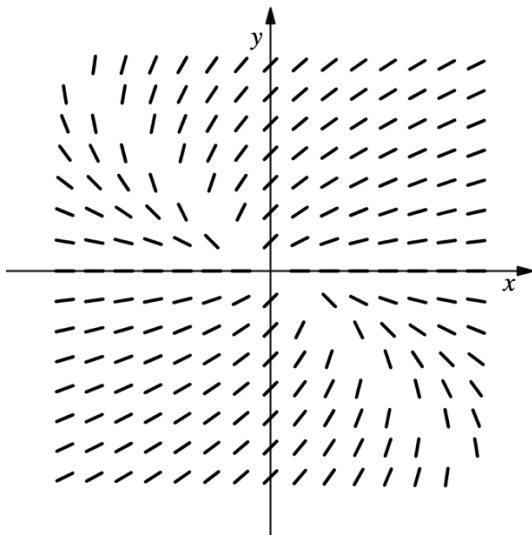
8 For non-zero vectors $\underline{p}$ and $\underline{q}$, what is $\text{proj}_{3\underline{q}} 6\underline{p}$ if $\text{proj}_{\underline{q}} \underline{p} = \underline{r}$?

- A. $\underline{r}$
- B. $2\underline{r}$
- C. $6\underline{r}$
- D. $18\underline{r}$

9 Find $\int \tan x \sec^2 x \, dx$.

- A. $\sec x + C$
- B. $\frac{1}{2} \tan^2 x + C$
- C. $\frac{\sin x}{\cos^3 x} + C$
- D. $\sec^2 x + C$

10 Which differential equation is a possible match with the slope field presented below?



- A. $y' = \frac{x}{x+y}$
- B. $y' = \frac{y}{x-y}$
- C. $y' = \frac{x}{x-y}$
- D. $y' = \frac{y}{x+y}$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section.

- Answer the questions in the spaces provided.
 - Your responses should include relevant mathematical reasoning and/or calculations.
-

Question 11 (15 marks) Use a separate Writing Booklet

- (a) Solve $\frac{4 - x^2}{x - 1} \leq 0$. 3
- (b) A committee of 7 is to be chosen from 6 men and 8 women. How many different committees can be formed if the committee must have exactly 4 men? 2
- (c) Three forces $F_1 = 80$ N at 290° T, $F_2 = 110$ N at 040° T and $F_3 = 90$ N at 180° T are acting on an object. Show that the resultant force $\vec{F}$ is $-4.5\mathbf{i} + 21.6\mathbf{j}$, corrected to one decimal place. 2
- (d) A binomial random variable, X , has $E(X) = \frac{2}{3}$ and $\text{Var}(X) = \frac{5}{9}$. Calculate $P(X \geq 1)$. 3
- (e) (i) Show that the equation $3 \cos x + 2 \sin x = -3$ can be written as $2t + 3 = 0$,
where $t = \tan \frac{x}{2}$. 2
- (ii) Hence solve the equation for $0 \leq x \leq 2\pi$, correct to 1 decimal place where necessary. 3

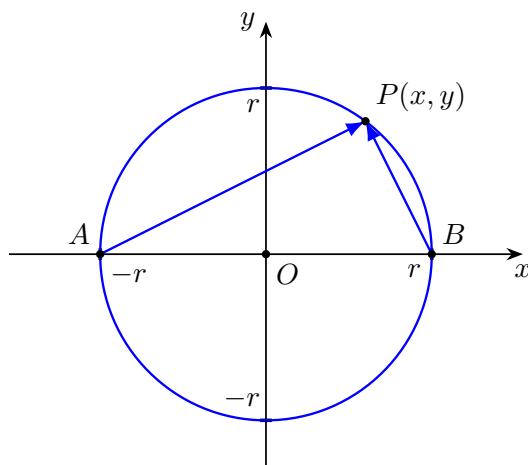
Question 12 (15 marks) Use a separate Writing Booklet

(a) Let $\underline{a} = \overrightarrow{OA}$ and $\underline{n} = \overrightarrow{ON}$. Find $\text{proj}_{\underline{n}} \underline{a}$ as a multiple of $\underline{n}$ if $|OA| = 4$, $|ON| = 6$ and $\angle AON = 30^\circ$. **2**

(b) Calculate the area of the region bounded by the curves $y = x^2 - 6$ and $y = x$, given that they intersect at $(3, 3)$ and $(-2, -2)$. **3**

(c) Find $\int \frac{\ln x}{x} dx$, using the substitution $u = \ln x$. **2**

(d) The graph of a circle with radius r and centre at $(0, 0)$ is shown below. **3**
Prove that $\overrightarrow{AP}$ and $\overrightarrow{BP}$ in the diagram are perpendicular using vectors.



(e) (i) Prove that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ using the expansion of $\cos(2\theta + \theta)$. **2**

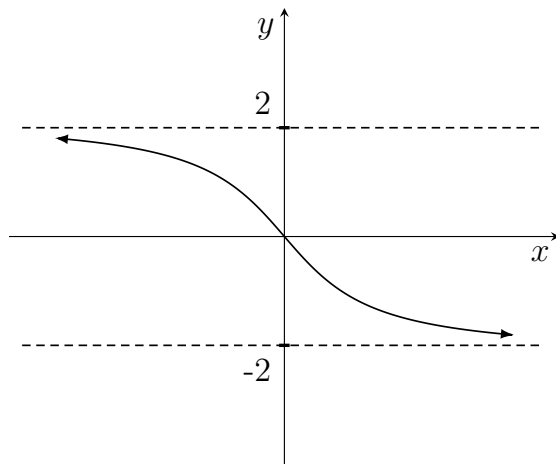
(ii) Hence solve $8x^3 - 6x - \sqrt{3} = 0$ using the substitution $x = \cos \theta$. **3**

Question 13 (15 marks) Use a separate Writing Booklet

- (a) (i) Explain why $y = \sin^{-1} x + \cos^{-1} x$ is a constant function. 2
- (ii) Hence find the constant. 2
- (b) Trevor states that $n^2 + 3n$ is an odd integer for all integers $n \geq 1$.
- (i) Show that the statement is true for $n = k + 1$ if it is true for $n = k$,
where k is an integer greater than or equal to 1. 2
- (ii) Is Trevor's statement true? Justify your answer. 2
- (c) A solid is formed by rotating the region bounded by the curve $y = 6 - x^2$ and $y = 2$
about the y -axis. Find the exact value of the volume of the solid. 3
- (d) Consider the differential equation $\frac{dy}{dx} = xe^{-y}$.
- (i) Explain why the differential equation does **not** have a constant solution. 2
- (ii) Find the other solutions of the differential equation by separating the variables. 2

Question 14 (15 marks) Use a separate Writing Booklet

- (a) Given the graph of $y = f(x)$ below, sketch $y = \frac{1}{f(x)}$, showing the key features clearly. 2



- (b) Prove by mathematical induction that $8(8^n - 1) - 7n$ is divisible by 49 for all integers $n \geq 0$. 4

- (c) A rabbit population of 500 was released on an island. The population growth is modelled by the logistic equation $\frac{dP}{dt} = \frac{P}{10} \left(1 - \frac{P}{2000} \right)$. 4

Given that $\frac{20000}{P(2000 - P)} = 10 \left(\frac{1}{P} + \frac{1}{2000 - P} \right)$, solve the differential equation

to show that the population P at time t months after introduction is $P = \frac{2000}{1 + 3e^{-\frac{t}{10}}}$.

- (d) A stone is projected from level ground with initial speed $V \text{ ms}^{-1}$ at an angle of θ to the horizontal. The maximum height reached by the stone was 8 metres.

- (i) By integrating vectors, show that the velocity and displacement of the stone at t seconds are as below. 2

$$\mathbf{v} = V \cos \theta \mathbf{i} + (V \sin \theta - gt) \mathbf{j}$$

$$\mathbf{r} = Vt \cos \theta \mathbf{i} + \left(Vt \sin \theta - \frac{g}{2} t^2 \right) \mathbf{j}$$

- (ii) Prove that the horizontal range of the stone is $\sqrt{\frac{64}{g}} (V^2 - 16g)$. 3

End of paper

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Calculus	4, 6, 9, 10 / 4		b, c / 3+2	a, c, d / 4+3+4	c / 4	/ 24
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1 Given that $\overrightarrow{OA} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$ and $\overrightarrow{OB} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$, What is the direction of $\overrightarrow{AB}$ to the nearest degree?

- A. 37°
- B. 53°
- C. -53°
- D. 143°

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$\overrightarrow{AB} = \begin{bmatrix} -3 \\ 4 \end{bmatrix} - \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$
$$= \begin{bmatrix} -8 \\ 6 \end{bmatrix}$$

$$\tan \theta = \frac{6}{8}$$

$$\theta = \tan^{-1} \frac{6}{8}$$

$$\theta = 37^\circ \text{ (nearest degree)}$$

$\therefore$ Since $(-8, 6)$ lies in the 2nd quadrant, the direction of $\overrightarrow{AB}$ is $180^\circ - 37^\circ = 143^\circ$.

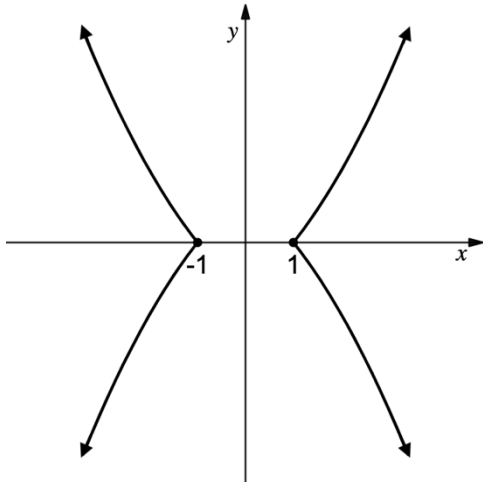
Answer: D

2 Mathematical induction is a method of proof that can be used to prove a statement:

- A. only for all positive integers n .
- B. for all integers $n \geq$ any fixed integer.
- C. for all real numbers n .
- D. for all integers n .

Answer: A or B

- 3 For $f(x) = (x + 3)(x - 1)$, which one of the following represents the graph below?



- A. $|y| = f(x)$
- B. $y = |f(|x|)|$
- C. $|y| = |f(x)|$
- D. $|y| = f(|x|)$

Answer: D

- 4 Which of the following integrals produces the volume of a cone?

- A. $\pi \int_0^2 x \, dx$
- B. $\pi \int_{-1}^2 (y^2 - 1)^2 \, dy$
- C. $\pi \int_{-1}^2 (2 - x)^2 \, dx$
- D. $\pi \int_{-2}^0 (2y - 2)^2 \, dy$

Answer: C

5 Find $P(X = 3)$ if $X \sim B\left(4, \frac{5}{8}\right)$.

A. $\binom{4}{3} \left(\frac{5}{8}\right)^3 \left(\frac{3}{8}\right)$

B. $1 - \binom{4}{3} \left(\frac{5}{8}\right) \left(\frac{3}{8}\right)^3$

C. $\left(\frac{5}{8}\right)^3$

D. $\binom{4}{3} \left(\frac{5}{8}\right) \left(\frac{3}{8}\right)^3$

Answer: A

6 Which of the following is **not** a first-order linear differential equation?

A. $xy' + x^2y = 1$

B. $y \sin x - y' \cos x = 0$

C. $y'y + 4x = 2$

D. $y' = 3x^2$

A first-order differential equation is called linear if it can be put into the form $y' + f(x)y = g(x)$, where $f(x)$ and $g(x)$ functions of x .

Option A is $y' + xy = \frac{1}{x}$

Option B is $y' - y \tan x = 0$

Option C can be rearranged as $y' + \frac{4x}{y} = \frac{2}{y}$ which is not in the required form.

Option D has the y term missing but it is still in the required form. It is a case when $f(x) = 0$.

Answer: C

- 7 A particle is projected with initial speed $V \text{ ms}^{-1}$ at an angle of α to the horizontal. If its position at t seconds is given by $\mathbf{r} = 40t\mathbf{i} + \left(9t - \frac{1}{2}gt^2\right)\mathbf{j}$, which of the following statements is correct?

- A. $V = 7 \text{ ms}^{-1}$
- B. $V = \frac{\sqrt{6481}}{2} \text{ ms}^{-1}$
- C. $V = \sqrt{1681 - g^2} \text{ ms}^{-1}$
- D. $V = 41 \text{ ms}^{-1}$

$$\mathbf{v} = 40\mathbf{i} + (9 - gt)\mathbf{j}$$

$$\text{When } t = 0, \quad \mathbf{v} = 40\mathbf{i} + 9\mathbf{j}$$

$$\begin{aligned} \text{Initial speed} &= \sqrt{40^2 + 9^2} \\ &= \sqrt{1681} \\ &= 41 \text{ ms}^{-1} \end{aligned}$$

Answer: D

- 8 For non-zero vectors $\mathbf{p}$ and $\mathbf{q}$, what is $\text{proj}_{3\mathbf{q}} 6\mathbf{p}$ if $\text{proj}_{\mathbf{q}} \mathbf{p} = \mathbf{r}$?

- A. $\mathbf{r}$
- B. $2\mathbf{r}$
- C. $6\mathbf{r}$
- D. $18\mathbf{r}$

Answer: C

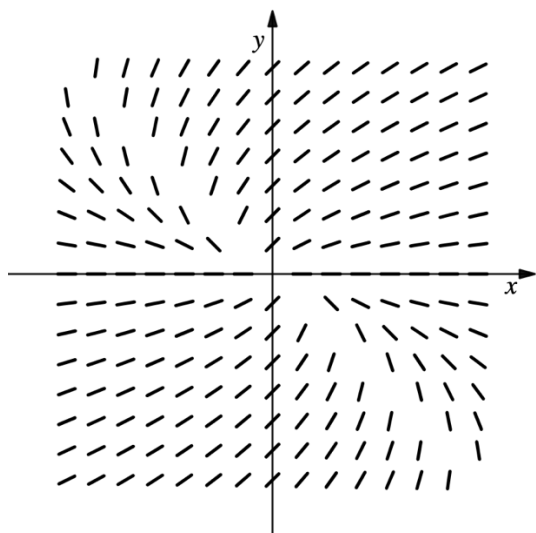
- 9 Find $\int \tan x \sec^2 x \, dx$.

- A. $\sec x + C$
- B. $\frac{1}{2}\tan^2 x + C$
- C. $\frac{\sin x}{\cos^3 x} + C$
- D. $\sec^2 x + C$

$$\begin{aligned} \int \sec^2 x (\tan x) \, dx &= \frac{(\tan x)^2}{2} + C && \text{(reverse chain rule)} \\ &= \frac{1}{2}\tan^2 x + C \end{aligned}$$

Answer: B

10 Which differential equation is a possible match with the slope field presented below?



A. $y' = \frac{x}{x+y}$

B. $y' = \frac{y}{x-y}$

C. $y' = \frac{x}{x-y}$

D. $y' = \frac{y}{x+y}$

- The slopes are undefined for the points where x and y values have the same magnitude but opposite signs. That is, when $x + y = 0$. This excludes option B and C.
- The slopes for the points on the x -axis are zero. That is, $y' = 0$ when $y = 0$. This excludes A.

Answer: D

Question 11 (15 marks) Use a separate Writing Booklet

(a) Solve $\frac{4-x^2}{x-1} \leq 0$.

3

$$\frac{4-x^2}{\cancel{x-1}} \times (x-1)^2 \leq 0 \times (x-1)^2, \quad \text{where } x \neq 1$$

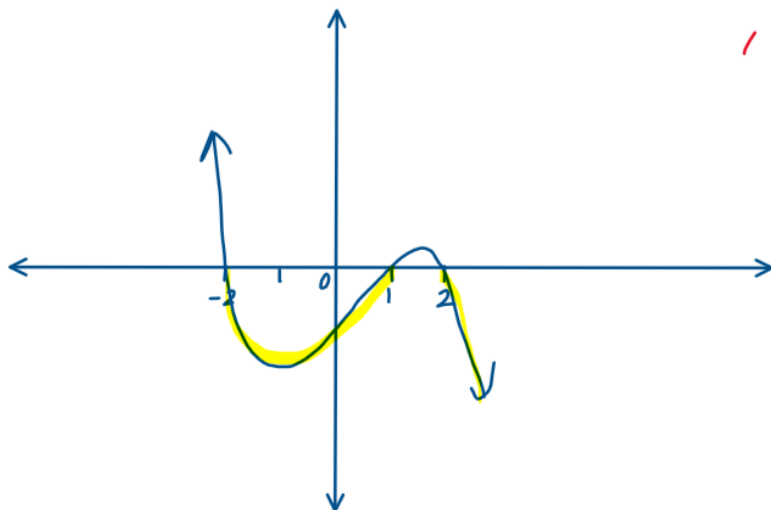
$$(4-x^2)(x-1) \leq 0$$

$$(2-x)(2+x)(x-1) \leq 0$$

1 mark for multiplying $(x-1)^2$ both sides

1 mark for $x \neq 1$

1 mark for correct solution



Comment: most students did not state $x \neq 1$.

$$\therefore -2 \leq x < 1 \quad \text{or } x \geq 2$$

(b) A committee of 7 is to be chosen from 6 men and 8 women. How many different committees can be formed if the committee must have exactly 4 men?

2

$${}^6C_4 \times {}^8C_3 = 840$$

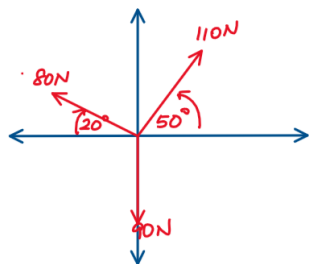
1 mark for each

2 marks for correct answer.

Well done

(c) Three forces $F_1 = 80 \text{ N}$ at 290°T , $F_2 = 110 \text{ N}$ at 040°T and $F_3 = 90 \text{ N}$ at 180°T are acting on an object. Show that the resultant force $\vec{F}$ is $-4.5\hat{i} + 21.6\hat{j}$, corrected to one decimal place.

2



$$\vec{F}_1 = -80 \cos 20^\circ \hat{i} + 80 \sin 20^\circ \hat{j}$$

$$\vec{F}_2 = 110 \cos 50^\circ \hat{i} + 110 \sin 50^\circ \hat{j}$$

$$\vec{F}_3 = -90 \hat{j}$$

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

$$= (-80 \cos 20^\circ + 110 \cos 50^\circ) \hat{i} + (80 \sin 20^\circ + 110 \sin 50^\circ - 90) \hat{j}$$

$$= -4.5 \hat{i} + 21.6 \hat{j}$$

Not well done.
Most students mixed sin and cos
and used 70° or 40° .

(d) A binomial random variable, X , has $E(X) = \frac{2}{3}$ and $\text{Var}(X) = \frac{5}{9}$. Calculate $P(X \geq 1)$.

3

$$E(X) = np = \frac{2}{3} \quad \text{--- (1)}$$

$$\text{Var}(X) = npq = \frac{5}{9} \quad \text{--- (2)}$$

Sub (1) into (2)

$$\frac{2}{3}q = \frac{5}{9}$$

$$q = \frac{5}{9} \times \frac{3}{2}$$
$$= \frac{5}{6}$$

$$p = 1 - q$$
$$= 1 - \frac{5}{6}$$

$$p = \frac{1}{6}$$

Sub $p = \frac{1}{6}$ into (1)

$$n\left(\frac{1}{6}\right) = \frac{2}{3}$$

$$n = \frac{2}{3} \times \frac{6}{1}$$

$$n = 4$$

$$P(X \geq 1) = 1 - P(X = 0)$$

$$= 1 - \left({}^4C_0 \times \left(\frac{1}{6}\right)^0 \times \left(\frac{5}{6}\right)^4 \right)$$

$$= 1 - \frac{625}{1296}$$

$$= \frac{671}{1296}$$

2 marks for solving simultaneously
and finding n, p, q .

1 mark for correct answer

- (e) (i) Show that the equation $3 \cos x + 2 \sin x = -3$ can be written as $2t + 3 = 0$,
where $t = \tan \frac{x}{2}$.

2

$$\frac{3(1-t^2)}{1+t^2} + \frac{2(2t)}{1+t^2} = -3$$

$$3 - 3t^2 + 4t = -3(1+t^2)$$

$$3 - \cancel{3t^2} + 4t = -3 - \cancel{3t^2}$$

$$4t + 6 = 0$$

$$2(2t + 3) = 0$$

$$2t + 3 = 0$$

Well done

- (ii) Hence solve the equation for $0 \leq x \leq 2\pi$, correct to 1 decimal place where necessary.

3

$$\begin{aligned} 2t + 3 &= 0 \\ 2t &= -3 \\ t &= -\frac{3}{2} \end{aligned}$$

$$\tan \frac{x}{2} = -\frac{3}{2} \quad \text{for } 0 \leq \frac{x}{2} \leq \pi$$

related angle $\frac{x}{2} = \tan^{-1}\left(\frac{3}{2}\right)$ $\tan$ is negative in quad 2.

$$\frac{x}{2} = 0.9827937232$$

1 mark for finding the related angle

Since $\tan \frac{x}{2} < 0$, $\frac{x}{2}$ lies in 2nd quadrant

$$\frac{x}{2} = \pi - 0.9827937232$$

$$= 2.15879893$$

$$\therefore x = 4.317597861$$

$$= 4.3 \text{ (1.d.p.)}$$

1 mark for 4.3

Check: when $x = \pi$ (since $t = \tan \frac{x}{2}$)

$$\text{LHS} = 3 \cos \pi + 2 \sin \pi$$

$$= -3 + 0$$

$$= -3$$

$$= \text{RHS}$$

$\therefore x = \pi$ is a solution

$\therefore x = 4.3 \text{ (1.d.p.) or } \pi$

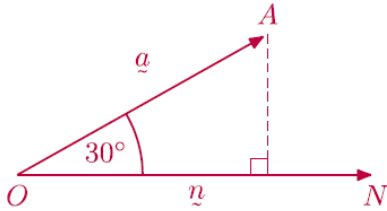
1 mark for checking $x = \pi$ and conclude it is a solution.

Question 12 (15 marks) Use a separate Writing Booklet

- (a) Let $\underline{a} = \overrightarrow{OA}$ and $\underline{n} = \overrightarrow{ON}$. Find $\text{proj}_{\underline{n}} \underline{a}$ as a multiple of $\underline{n}$ if $|OA| = 4$, $|ON| = 6$ and $\angle AON = 30^\circ$.

2

Solution 1:



$$\begin{aligned} \text{proj}_{\underline{n}} \underline{a} &= (|\underline{a}| \cos 30^\circ) \hat{n} \Rightarrow \text{1 mark} \\ &= 4 \times \frac{\sqrt{3}}{2} \times \frac{\underline{n}}{6} \\ &= \frac{\sqrt{3}}{3} \underline{n} \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{proj}_{\underline{n}} \underline{a} &= (|\underline{a}| \cos 30^\circ) \hat{n} \\ &= 4 \times \frac{\sqrt{3}}{2} \times \frac{\underline{n}}{6} \\ &= \frac{\sqrt{3}}{3} \underline{n} \end{aligned}} \right\} \text{1 mark}$$

Solution 2:

$$\begin{aligned} \underline{a} &= \begin{bmatrix} 4 \cos 30^\circ \\ 4 \sin 30^\circ \end{bmatrix}, & \underline{n} &= \begin{bmatrix} 6 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 2\sqrt{3} \\ 2 \end{bmatrix} \\ \text{proj}_{\underline{n}} \underline{a} &= \frac{\underline{a} \cdot \underline{n}}{\underline{n} \cdot \underline{n}} \underline{n} \\ &= \frac{12\sqrt{3} + 0}{36 + 0} \underline{n} \\ &= \frac{\sqrt{3}}{3} \underline{n} \end{aligned}$$

Marker's Comment:

Common error is not using the correct formula for the projection formula. Some students used this incorrect formula with the wrong denominator:

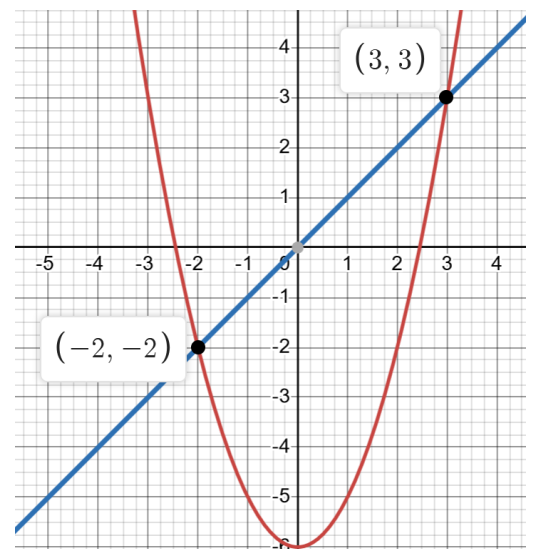
$$\text{proj}_{\underline{n}} \underline{a} = \frac{\underline{a} \cdot \underline{n}}{\underline{a} \cdot \underline{a}} \underline{n}$$

However, this question was mostly well done.

- (b) Calculate the area of the region bounded by the curves $y = x^2 - 6$ and $y = x$, given that they intersect at $(3, 3)$ and $(-2, -2)$.

3

$$\begin{aligned} A &= \int_{-2}^3 (x - (x^2 - 6)) dx \Rightarrow \text{1 mark} \\ &= \int_{-2}^3 (x - x^2 + 6) dx \\ &= \left[\frac{x^2}{2} - \frac{x^3}{3} + 6x \right]_{-2}^3 \Rightarrow \text{1 mark} \\ &= \left(\frac{3^2}{2} - \frac{3^3}{3} + 6(3) \right) - \left(\frac{(-2)^2}{2} - \frac{(-2)^3}{3} + 6(-2) \right) \Rightarrow \text{1 mark} \\ &= \frac{125}{6} \text{ unit}^2 \end{aligned}$$



Marker's Comment:

It is important that student's show the line of substitution. Always

show the substitution of the upper limit and lower limit. You may be awarded marks even though you made a calculation error on your calculator.

This question was mostly well done.

(c) Find $\int \frac{\ln x}{x} dx$, using the substitution $u = \ln x$.

2

$$\frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{1}{x} dx$$

$$\begin{aligned} \int \frac{\ln x}{x} dx &= \int \frac{1}{x} \ln x dx \\ &= \int u du \\ &= \frac{u^2}{2} + C \\ &= \frac{1}{2} (\ln x)^2 + C \end{aligned} \quad \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \text{1 mark} \\ \\ \Rightarrow \text{1 mark} \end{array}$$

Marker's Comment:

Most common error is interpreting $(\ln x)^2$ as $\ln x^2$.

$$(\ln x)^2 = (\ln x)(\ln x)$$

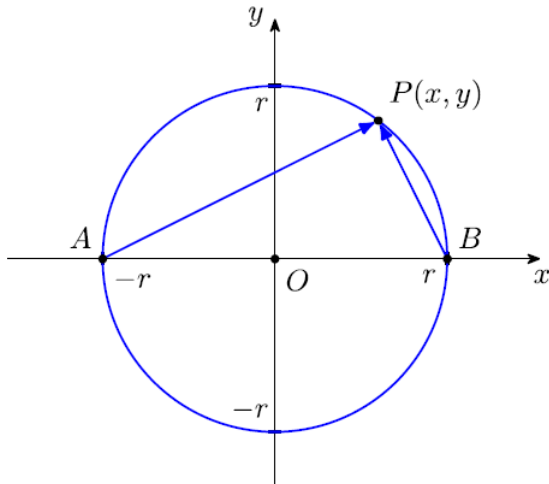
$$\ln x^2 = 2 \ln x$$

Some students forgot to substitute back the $\ln x$ on to u .

However, this question was mostly well done.

(d) The graph of a circle with radius r and centre at $(0, 0)$ is shown below.

Prove that $\overrightarrow{AP}$ and $\overrightarrow{BP}$ in the diagram are perpendicular using vectors.



Required to show: $\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$ ($\cos 90^\circ = 0$)

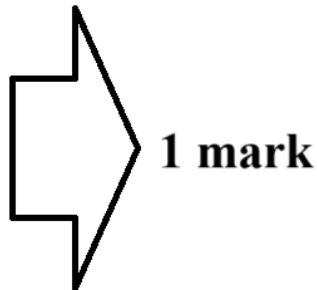
$$\begin{aligned}\overrightarrow{AP} &= \overrightarrow{OP} - \overrightarrow{OA} \\ &= \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} -r \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} x+r \\ y \end{bmatrix}\end{aligned}$$

$$\begin{aligned}\overrightarrow{BP} &= \overrightarrow{OP} - \overrightarrow{OB} \\ &= \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} r \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} x-r \\ y \end{bmatrix}\end{aligned}$$

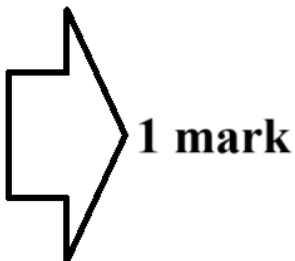
$$\begin{aligned}\overrightarrow{AP} \cdot \overrightarrow{BP} &= \begin{bmatrix} x+r \\ y \end{bmatrix} \cdot \begin{bmatrix} x-r \\ y \end{bmatrix} \\ &= (x+r)(x-r) + y^2 \\ &= x^2 - r^2 + y^2 \\ &= x^2 + y^2 - r^2 \\ &= r^2 - r^2 \\ &= 0\end{aligned}$$

Since $\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$

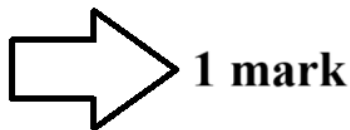
$\therefore \overrightarrow{AP}$ and $\overrightarrow{BP}$ are perpendicular.



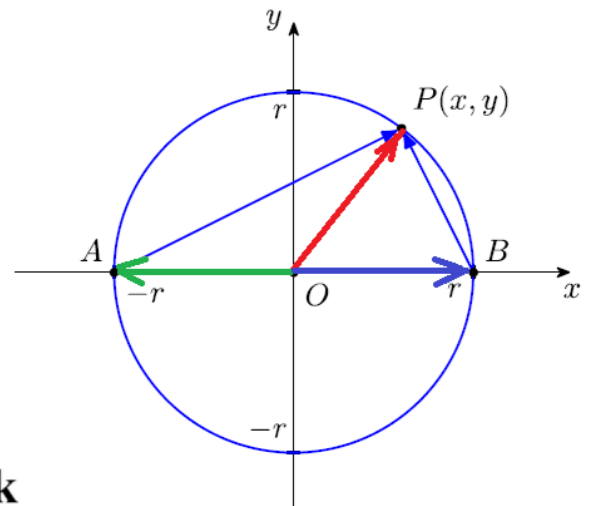
1 mark



1 mark



1 mark



Marker's Comment:

No marks are awarded for students who did not use vectors to prove that AP and BP are perpendicular.

So far, this section was not too bad.

(e) (i) Prove that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ using the expansion of $\cos(2\theta + \theta)$.

2

$$\begin{aligned}\text{LHS} &= \cos 3\theta \\ &= \cos(2\theta + \theta) \\ &= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\ &= (2\cos^2 \theta - 1)\cos \theta - 2\sin^2 \theta \cos \theta \\ &= 2\cos^3 \theta - \cos \theta - 2\cos \theta (1 - \cos^2 \theta) \\ &= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta \\ &= 4\cos^3 \theta - 3\cos \theta \\ &= \text{RHS}\end{aligned}$$


1 mark

1 mark

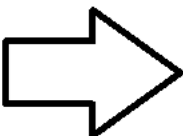
Marker's Comment:

Mostly well done.

(ii) Hence solve $8x^3 - 6x - \sqrt{3} = 0$ using the substitution $x = \cos \theta$.

3

Substitute $x = \cos \theta$,

$$\begin{aligned}8\cos^3 \theta - 6\cos \theta - \sqrt{3} &= 0 \\ 8\cos^3 \theta - 6\cos \theta &= \sqrt{3} \\ 4\cos^3 \theta - 3\cos \theta &= \frac{\sqrt{3}}{2} \\ \cos 3\theta &= \frac{\sqrt{3}}{2} \quad (\text{from (i)})\end{aligned}$$


1 mark

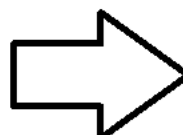
Related angle $= \cos^{-1} \frac{\sqrt{3}}{2}$

$$= \frac{\pi}{6}$$

Since $\cos 3\theta > 0$, 3θ lies in the 1st and 4th quadrants.

$$3\theta = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6}, \frac{25\pi}{6}, \dots$$


1 mark

$$\theta = \frac{\pi}{18}, \frac{11\pi}{18}, \frac{13\pi}{18}, \frac{23\pi}{18}, \frac{25\pi}{18}, \dots$$
$$x = \cos \frac{\pi}{18}, \cos \frac{11\pi}{18}, \cos \frac{13\pi}{18}$$


1 mark

Marker's Comment:

Most students lost mark by not listing more than 3 values. It is important that students list more than 3 although it is a cubic polynomial since we are looking for distinct roots and verify that they are distinct values. Some students forgot to solve for x eventually, thus losing marks again.

Question 13 (15 marks) Use a separate Writing Booklet

(a) (i) Explain why $y = \sin^{-1} x + \cos^{-1} x$ is a constant function.

2

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0 \quad 1 \text{ mark}$$

$\therefore$ The function is constant since its derivative is zero. 1 mark

If you have used a graphical method, you must provide a detailed explanation to earn 2 marks.

(ii) **Hence** find the constant.

2

$$\sin^{-1} x + \cos^{-1} x = c \quad (\text{from (a)})$$

Substitute $x = 0$, (since y is constant, choose any x value from the domain of $-1 \leq x \leq 1$)

$$c = \sin^{-1} 0 + \cos^{-1} 0 = 0 + \frac{\pi}{2} \quad 1 \text{ mark}$$

$$\therefore c = \frac{\pi}{2} \quad 1 \text{ mark}$$

'Hence ...'
Penalised for any other methods.

(b) Trevor states that $n^2 + 3n$ is an odd integer for all integers $n \geq 1$.

(i) Show that the statement is true for $n = k + 1$ if it is true for $n = k$, where k is an integer greater than or equal to 1.

2

Assume true for $n = k$, where k is an integer.

i. e. $k^2 + 3k = 2P + 1$, where P is an integer.

Prove true for $n = k + 1$.

i. e. $(k + 1)^2 + 3(k + 1) = 2Q + 1$, where Q is an integer.

$$\begin{aligned} \text{LHS} &= k^2 + 2k + 1 + 3k + 3 \\ &= (2P + 1 - 3k) + 5k + 4 \quad (\text{by the assumption}) \\ &= 2P + 2k + 5 \\ &= 2(P + k + 2) + 1 \\ &= 2Q + 1 \\ &= \text{RHS} \end{aligned}$$

$\therefore$ The statement is true for $n = k + 1$ if it is true for $n = k$. 1 mark

1 mark

Most common error:
 $3(k + 1) \neq 3k + 1$

(ii) Is Trevor's statement true? Justify your answer.

2

Prove true for $n = 1$,

$$1^2 + 3(1) = 4, \text{ which is not odd.} \quad 1 \text{ mark}$$

Since Trevor's statement is false for $n = 1$, his statement of not true. 1 mark

Must answer the questions to get the 2nd mark.

Question 13 (continues)

- (c) A solid is formed by rotating the region bounded by the curve $y = 6 - x^2$ and $y = 2$ about the y -axis. Find the exact value of the volume of the solid.

3

$$x^2 = 6 - y$$

$$V = \pi \int_2^6 x^2 dy$$

$$= \pi \int_2^6 (6 - y) dy \quad 1 \text{ mark}$$

$$= \pi \left[6y - \frac{y^2}{2} \right]_2^6$$

$$= \pi \left[\left(6(6) - \frac{6^2}{2} \right) - \left(6(2) - \frac{2^2}{2} \right) \right] \quad 1 \text{ mark}$$

$$= 8\pi \text{ unit}^3 \quad 1 \text{ mark}$$

Poorly done.

Common errors:

- Missing π .
- Incorrect limits.
- Integrating $(6 - y)^2$

The substitution step must be shown.
'Fundamental theorem of calculus.'

- (d) Consider the differential equation $\frac{dy}{dx} = xe^{-y}$.

- (i) Explain why the differential equation does **not** have a constant solution.

2

Substitute $\frac{dy}{dx} = 0$ for the constant solution,

$$0 = xe^{-y}$$

$$e^{-y} = 0$$

$\therefore y$ is undefined since e^{-y} cannot be zero.

$\therefore$ The constant solution does not exist.

1 mark

1 mark

- (ii) Find the other solutions of the differential equation by separating the variables.

2

$$\frac{dy}{dx} = xe^{-y}$$

$$e^y dy = x dx$$

Integrating both sides,

$$\int e^y dy = \int x dx$$

$$e^y = \frac{x^2}{2} + C$$

$$y = \ln \left| \frac{x^2}{2} + C \right|$$

1 mark

1 mark

Common errors:

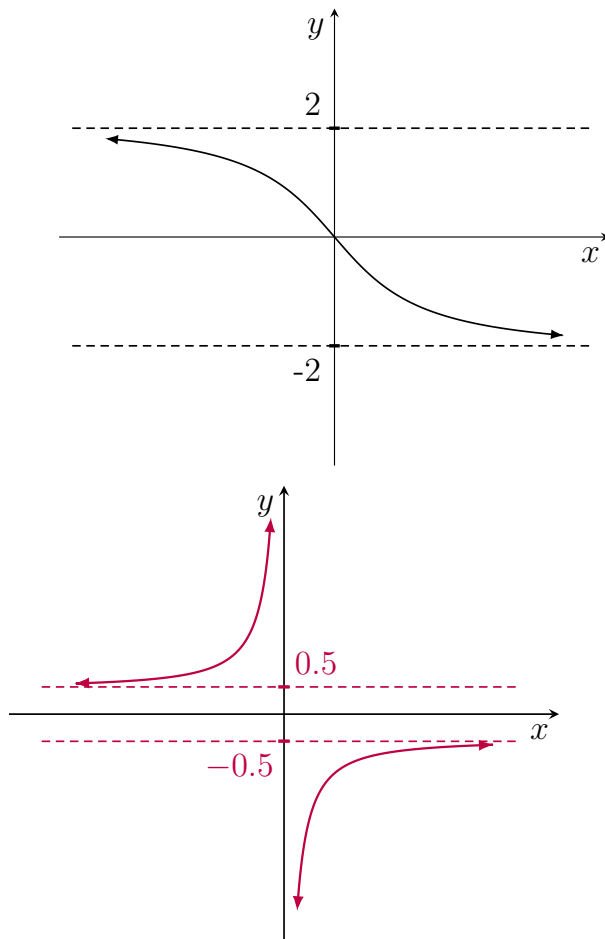
$$\ln \left(\frac{x^2}{2} + C \right) \neq \ln \left(\frac{x^2}{2} \right) + C$$

$$\neq \ln \left(\frac{x^2}{2} \right) + \ln C$$

2024 Feedback – Question 14

Question 14 (15 marks) Use a separate Writing Booklet

- (a) Given the graph of $y = f(x)$ below, sketch $y = \frac{1}{f(x)}$, showing the key features clearly. 2



Common Mistakes:

- Some students wrote incorrect equations for the asymptotes.
- Some students drew the graph within the asymptotes
- Overall, well done

Learning Strategies:

Students need to

- use a ruler to draw number plane,
- sketch smooth curves,
- pay attention to the key features such as x-intercepts, y-intercepts, asymptotes etc.
- Always label a coordinate on the curve

(b) Prove by mathematical induction that $8(8^n - 1) - 7n$ is divisible by 49 for all integers $n \geq 0$. 4

Step 1 Prove true for $n = 0$.

$8(8^0 - 1) - 0 = 0$, which is divisible by 49.

$\therefore$ The statement is true for $n = 0$.

Step 2 Assume true for $n = k$, where k is an integer $k \geq 0$.

i.e. assume $8(8^k - 1) - 7k = 49P$, where P is an integer.

$$8(8^k) - 8 - 7k = 49P$$

$$8(8^k) = 49P + 7k + 8$$

Now prove true for $n = k + 1$.

i.e. prove $8(8^{k+1} - 1) - 7(k + 1)$ is divisible by 49.

$$\text{LHS} = 8(8^{k+1} - 1) - 7(k + 1)$$

$$= 8(8^k \cdot 8 - 1) - 7k - 7$$

$$= 8(49P + 7k + 8 - 1) - 7k - 7 \quad (\text{by the assumption})$$

$$= 8(49P) + 56k + 56 - 7k - 7$$

$$= 8(49P) + 49k + 49$$

$$= 49(8P + k + 1), \text{ which is divisible by 49.}$$

Step 3 By the principle of mathematical induction, the statement is true for all integers $n \geq 0$.

Common mistakes:

- Some students forgot to show the proof for the base value of n ($n = 0$), therefore lost one mark
- Very few students did the proof for divisible by 7 instead of 49
- Very few showed incorrect working for the $n = k + 1$ case, did not write using assumption, hence lost a mark
- Few students used the assumption but made the expression complicated
- Few students forgot to mention that the expression with 49 is an integer, hence lost a mark
-

Overall well done.

Learning Strategies:

To attempt questions on mathematical induction, students need to

- Follow teachers notes closely and write all the four steps in the exam

Question 14 (continues)

(c) A rabbit population of 500 was released on an island. The population growth is

4

modelled by the logistic equation $\frac{dP}{dt} = \frac{P}{10} \left(1 - \frac{P}{2000} \right)$.

Given that $\frac{20000}{P(2000 - P)} = 10 \left(\frac{1}{P} + \frac{1}{2000 - P} \right)$, solve the differential equation

to show that the population P at time t months after introduction is $P = \frac{2000}{1 + 3e^{-\frac{t}{10}}}$.

$$\frac{dP}{dt} = \frac{P}{10} \left(1 - \frac{P}{2000} \right)$$

$$\frac{dP}{dt} = \frac{P}{10} \left(\frac{2000 - P}{2000} \right)$$

$$\frac{dP}{dt} = \frac{P(2000 - P)}{20000}$$

$$\frac{20000}{P(2000 - P)} dP = dt$$

$$\int \frac{20000}{P(2000 - P)} dP = \int dt$$

$$10 \int \left(\frac{1}{P} + \frac{1}{2000 - P} \right) dP = \int dt$$

$$\ln|P| - \ln|2000 - P| = \frac{t}{10} + C$$

Majority of the students arrived at the above step. Well done.

$$\ln \left| \frac{P}{2000 - P} \right| = \frac{t}{10} + C$$

$$\left| \frac{P}{2000 - P} \right| = e^{\frac{t}{10} + C}$$

$$\frac{P}{2000 - P} = \pm e^C e^{\frac{t}{10}}$$

$$P = Ae^{\frac{t}{10}}(2000 - P), \quad \text{where } A \text{ is a nonzero constant}$$

$$P = 2000Ae^{\frac{t}{10}} - PAe^{\frac{t}{10}}$$

$$P(1 + Ae^{\frac{t}{10}}) = 2000Ae^{\frac{t}{10}}$$

$$P = \frac{2000Ae^{\frac{t}{10}}}{1 + Ae^{\frac{t}{10}}}$$

Dividing by $Ae^{\frac{t}{10}}$,

$$P = \frac{2000}{\frac{1}{A}e^{-\frac{t}{10}} + 1}$$

$$P = \frac{2000}{Be^{-\frac{t}{10}} + 1} \text{ where } B \text{ is a nonzero constant}$$

Substitute $t = 0, P = 500$,

$$500 = \frac{2000}{B + 1}$$

$$1 + B = 4$$

$$\therefore B = 3$$

$$\therefore P = \frac{2000}{1 + 3e^{-\frac{t}{10}}}$$

- few students messed up the expression and did not arrive at the desired step

- few students stopped just one line before the desired expression, I could not award them as it was a “show that” question

Overall well done.

Learning Strategies:

For Logistics questions, try to avoid the bounds on the integration, stick to finding the value of the constant using the given conditions.

Question 14 (continues)

(d) A stone is projected from level ground with initial speed $V \text{ ms}^{-1}$ at an angle of θ to the horizontal. The maximum height reached by the stone was 8 metres.

(i) By integrating vectors, show that the velocity and displacement of the stone at t seconds are as below.

2

$$\underline{v} = V \cos \theta \mathbf{i} + (V \sin \theta - gt) \mathbf{j}$$

$$\underline{r} = Vt \cos \theta \mathbf{i} + \left(Vt \sin \theta - \frac{g}{2} t^2 \right) \mathbf{j}$$

$$\underline{a} = -g \mathbf{j}$$

$$\underline{v} = \int -g \mathbf{j} dt$$

$$\underline{v} = -gt \mathbf{j} + \underline{C}$$

$$\text{When } t = 0, \underline{v} = V \cos \theta \mathbf{i} + V \sin \theta \mathbf{j},$$

$$\underline{C} = V \cos \theta \mathbf{i} + V \sin \theta \mathbf{j}$$

$$\underline{v} = -gt \mathbf{j} + V \cos \theta \mathbf{i} + V \sin \theta \mathbf{j}$$

$$\therefore \underline{v} = V \cos \theta \mathbf{i} + (V \sin \theta - gt) \mathbf{j}$$

$$\underline{r} = \int (V \cos \theta \mathbf{i} + (V \sin \theta - gt) \mathbf{j}) dt$$

$$\underline{r} = Vt \cos \theta \mathbf{i} + \left(Vt \sin \theta - \frac{gt^2}{2} \right) \mathbf{j} + \underline{D}$$

$$\text{When } t = 0, \underline{r} = \underline{0},$$

$$\underline{D} = \underline{0}$$

$$\therefore \underline{r} = Vt \cos \theta \mathbf{i} + \left(Vt \sin \theta - \frac{gt^2}{2} \right) \mathbf{j}$$

- Many students derived the equations for displacement vector and the velocity vector well
- Some students just wrote the integration without finding the values of the constant and since it is a "show that" question, they lost mark(s).

Overall, well done.

(ii) Prove that the horizontal range of the stone is $\sqrt{\frac{64}{g}}(V^2 - 16g)$.

3

Maximum height of 8 when $\dot{y} = 0$

$$V \sin \theta - gt = 0$$

$$t = \frac{V \sin \theta}{g} \quad (1)$$

- Many students arrived at this step successfully.

$$y = Vt \sin \theta - \frac{gt^2}{2}$$

$$8 = V \sin \theta \left(\frac{V \sin \theta}{g} \right) - \frac{g}{2} \left(\frac{V \sin \theta}{g} \right)^2$$

$$8 = \frac{V^2 \sin^2 \theta}{g} - \frac{V^2 \sin^2 \theta}{2g}$$

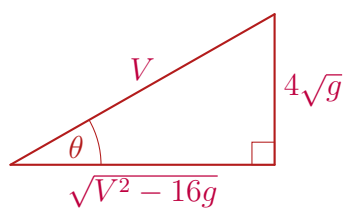
$$16g = 2V^2 \sin^2 \theta - V^2 \sin^2 \theta$$

$$16g = V^2 \sin^2 \theta$$

$$\sin^2 \theta = \frac{16g}{V^2}$$

$$\sin \theta = \frac{4\sqrt{g}}{V} \quad (\sin \theta > 0 \text{ since } \theta \text{ is acute})$$

- Some students arrived at this step, but did algebraic mistakes, and as a result, only few could arrive at this step



$$\therefore \cos \theta = \frac{\sqrt{V^2 - 16g}}{V}$$

$$\begin{aligned} \text{Time of flight} &= \frac{2V \sin \theta}{g} && (\text{From (1)}) \\ &= \frac{8\sqrt{g}}{g} \\ &= \frac{8}{\sqrt{g}} \end{aligned}$$

$$\begin{aligned} x &= Vt \cos \theta \\ &= V \left(\frac{8}{\sqrt{g}} \right) \left(\frac{\sqrt{V^2 - 16g}}{V} \right) \\ &= \frac{8\sqrt{V^2 - 16g}}{\sqrt{g}} \\ &= \sqrt{\frac{64(V^2 - 16g)}{g}} \\ \therefore x &= \sqrt{\frac{64}{g}(V^2 - 16g)} \end{aligned}$$

- Very few students arrived at this step.
- Poor attempt
-

Learning Strategies:

Students are advised to challenge themselves with questions which requires an in depth knowledge of the topic(s)